The 5-step PSA procedure is a systematic and organized method of solving any single phase or three phase power system analysis problem. This special procedure breaks up a power system analysis problem into an orderly sequence of five logical steps which are consecutively and separately done. The 5-step PSA procedure allows both small & simple and very large & complicated power system analysis problems to be methodically solved in exactly the same way. The five steps of the PSA procedure are explained in detail below.

1. **Visualize** the power system by reading the problem and gaining an understanding of what specific components are present in the power system and how these components connected together to create the power system. Then draw either a block diagram or one line diagram for the power system. If a block diagram or one line diagram is given to you with the problem, then you should not redraw it.

2. Draw an **Equivalent Circuit** which mathematically models the power system. The standard method of drawing power system equivalent circuits must be used, with the reference node line at the very bottom of the circuit and nothing drawn below this line. All nodes in the equivalent circuit should be labeled. All voltage sources, impedances, and admittances in the circuit must be clearly labeled with proper complex variable names and notation. Remember that if the power system is three phase, then you are drawing a L-N equivalent (or per phase) circuit. Do not do any calculations at all in this step.

3. Compute the equivalent **Circuit Parameters** from the information given in the power system specifications. The standard rules, methods, and equations defining each power system component must be used to perform these calculations. The values of all voltage sources, impedances, and admittances in the circuit must be calculated and the same complex variable names used in the equivalent circuit must be used in these calculations. Remember that if the power system is three phase, then you are calculating L-N equivalent (or per phase) circuit parameters, and proper interpretation and conversion of the specified system data must be made. Do not do any circuit analysis at all in this step.

4. Perform a **Circuit Analysis** of the equivalent circuit to obtain all of the desired equivalent circuit voltage, current, and power quantities needed for final results. Any circuit analysis methods can be used, but try to choose the method that is easiest and requires the least amount of work for the given circuit. Remember that all quantities calculated in this step are equivalent circuit quantities, not actual power system quantities. This is an especially important concept to remember in three phase power systems, which are modeled and analyzed using L-N equivalent (or per phase) circuits in which all voltages are L-N equivalent voltages, all currents are equivalent line currents, and all powers are L-N equivalent powers.

5. Present the **Final Results** for the actual power system that are derived from the equivalent circuit analysis results. Some minor calculations may be required here, but no additional circuit analysis should be done in this step. In single phase systems, if asked for voltage or current, then magnitude only results should be presented. In three phase systems, if asked for voltage or current, then L-L voltage magnitude or line current magnitude must be presented unless specific phasor quantities are asked for. If asked for powers in a three phase system, then three phase powers must be presented.

When using the 5-step PSA procedure, you must always label each of the 5-steps in your work and isolate these 5-steps from each other in your work. The following short hand labeling scheme should be used to label each of the 5-steps in your work.

Step 1: Visualize
Step 2: Equivalent Circuit
Step 3: Circuit Parameters
Step 4: Circuit Analysis
Step 5: Final Results

Do not label the steps with just 1, 2, 3, 4, 5 since that is not clear and could easily be confused with problem numbers in an assignment or test. The 5-step PSA procedure should be used in a precise and formal manner.